Report_DEB

1. Importation des données

```
file_path <- here::here("mod/Correction_X/A_DEB_SAAWKe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```
NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
seeds <- c(C1 = 3333, C2 = 3663, C3 = 2212)

text_priors <- '
Distrib (Fm,           TruncNormal,         0.1,      0.2,  0.0001,      0.8);
Distrib (pAm,          TruncNormal_cv,      100,      0.2,    40,    2000);
Distrib (r_pAm_pM,     TruncNormal_cv,      0.8,      0.1,    0.1,      2);
Distrib (v,            TruncNormal_cv,  0.18505,      0.1,  0.001,    2);
Distrib (kap,          Uniform,             0,     1);
Distrib (kapA,         Uniform,             0.1,     0.9);
Distrib (Ehp,          TruncNormal_cv,       100,      0.2,    40,    300);
Distrib (E_coc,        TruncNormal_cv,        30,      0.2,     1,    300);
Distrib (rOM_ClxHorse, LogUniform,       0.00001,      0.3);
Distrib (K,            Uniform,          0.00001,        2);
Distrib (dv,           TruncNormal_cv,         2,      0.5,     0,     10);
Distrib (wE,           LogUniform,     0.000000001,     0.1);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIte

# ./mcsim.DEBcali DEB_C1.in
```

`OM_horse` and `OM_soil` distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

```
                  Fm.1.     pAm.1. r_pAm_pM.1.       v.1.     kap.1.    kapA.1.
```

```
Result_mode 0.002097230 169.514000  0.76399400 0.18100800 0.92502700 0.50142900
2.5%        0.000624619 150.836150  0.67099500 0.15705900 0.87407400 0.43670600
97.5%       0.005379820 188.487000  0.88223555 0.22715000 0.94919800 0.58740000
Min         0.000460507  94.587600  0.62550900 0.13416000 0.50515300 0.18063900
Max         0.346466000 201.568000  0.96084700 0.25017900 0.95506900 0.73462200
Result_mean 0.002540534 168.828950  0.77842643 0.19080751 0.91428156 0.50767960
Result_sd   0.003035418   9.532371  0.05652069 0.01776346 0.01902079 0.03931835
               Ehp.1. E_coc.1. rOM_ClxHorse.1.       K.1.     dv.1.
Result_mode  95.50160 29.61690    0.0090942000 0.01876700 4.3031000
2.5%         74.64180 23.50445    0.0059353900 0.00558501 3.3081990
97.5%       143.43700 36.92140    0.0114057500 0.04290120 5.4187300
Min          62.73610 20.34090    0.0000161998 0.00371315 0.5220400
Max         169.85000 44.15910    0.0154621000 0.98193800 6.8222400
Result_mean 106.53838 29.44445    0.0086790930 0.02063286 4.3592407
Result_sd    19.39631  3.38544    0.0014252753 0.01224471 0.5476756
                   wE.1. Sigma_W.1.
Result_mode 1.988910e-09 0.12917800
2.5%        1.207650e-09 0.11788200
97.5%       3.076634e-06 0.15003300
Min         1.012790e-09 0.10613800
Max         7.044090e-06 5.72094000
Result_mean 4.066306e-07 0.13322549
Result_sd   8.414727e-07 0.05371061

Number of observations, n =  277
Number of parameters, k =  12
Log-likelihood, LL =  -33.13636
BIC =  133.7609


Potential scale reduction factors:

                Point est. Upper C.I.
dv.1.                 1.27       1.77
E_coc.1.              1.44       2.14
Ehp.1.                1.07       1.23
Fm.1.                 2.29       4.12
K.1.                  1.91       3.46
kap.1.                1.04       1.04
kapA.1.               1.02       1.05
pAm.1.                1.09       1.28
r_pAm_pM.1.           1.07       1.24
```

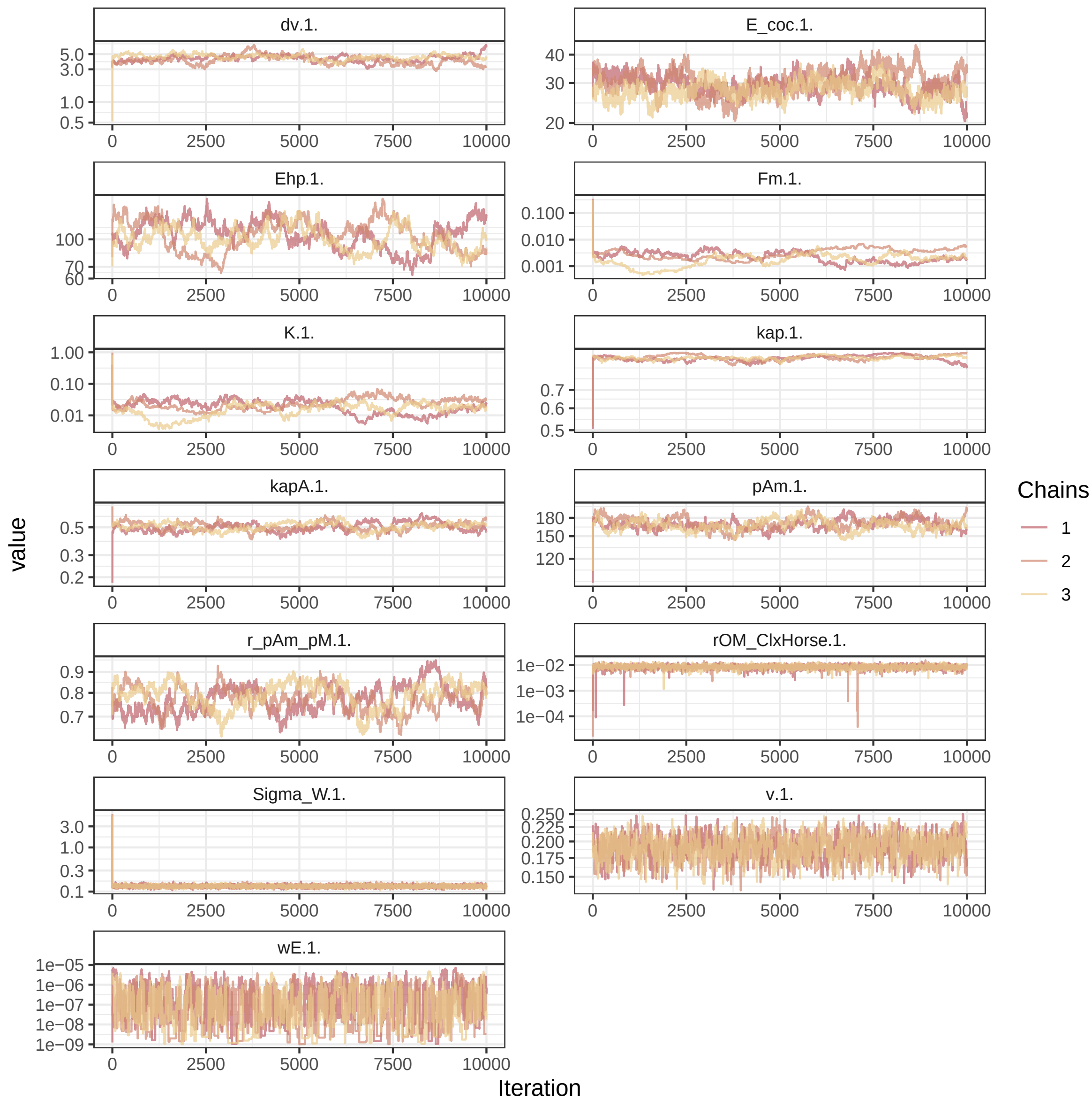


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

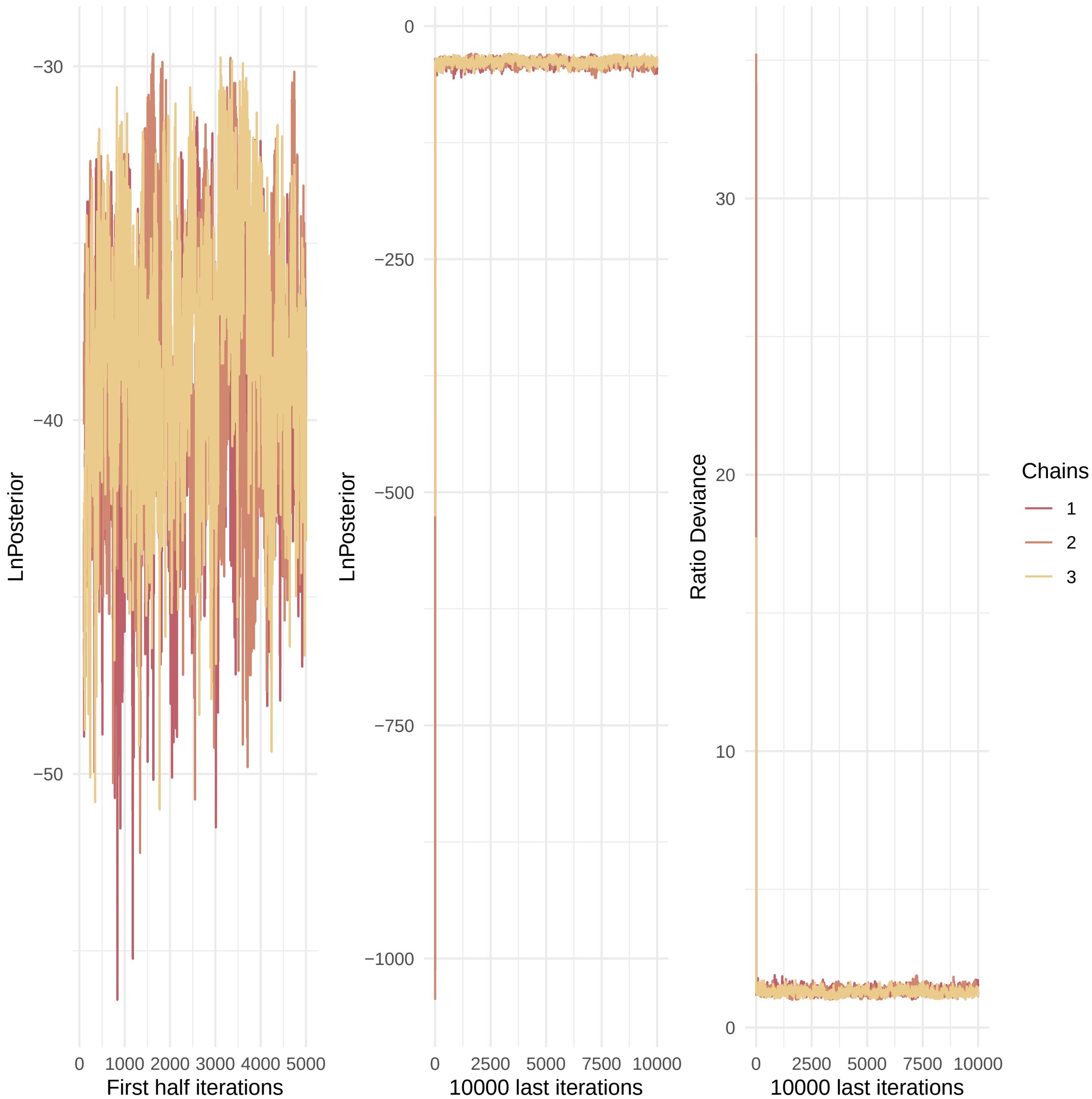


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

```
rOM_ClxHorse.1.       1.00       1.01
Sigma_W.1.            1.01       1.03
v.1.                  1.00       1.00
wE.1.                 1.07       1.13

Multivariate psrf

1.84
```

```r
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

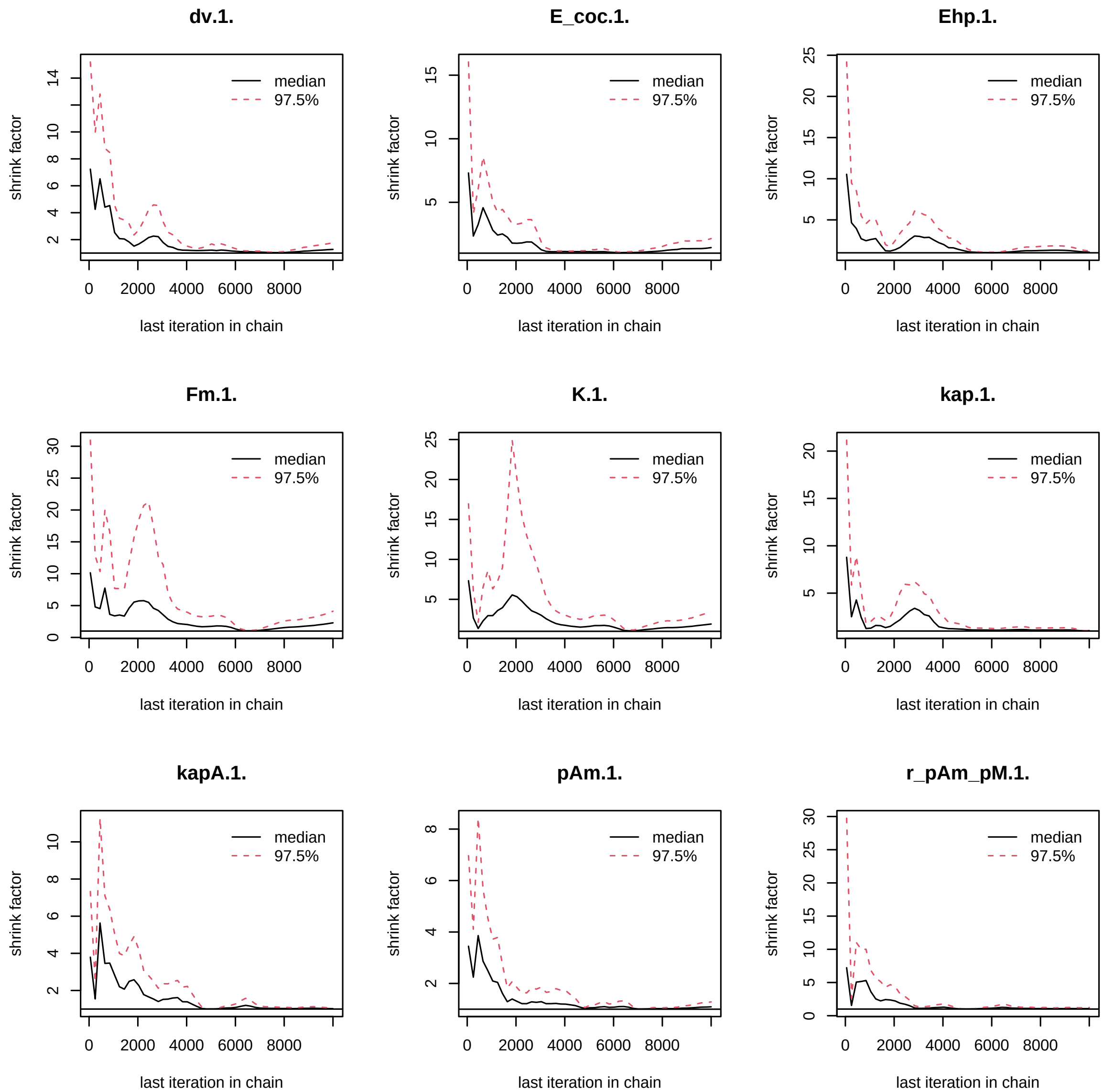


Figure 3: Chains convergence.

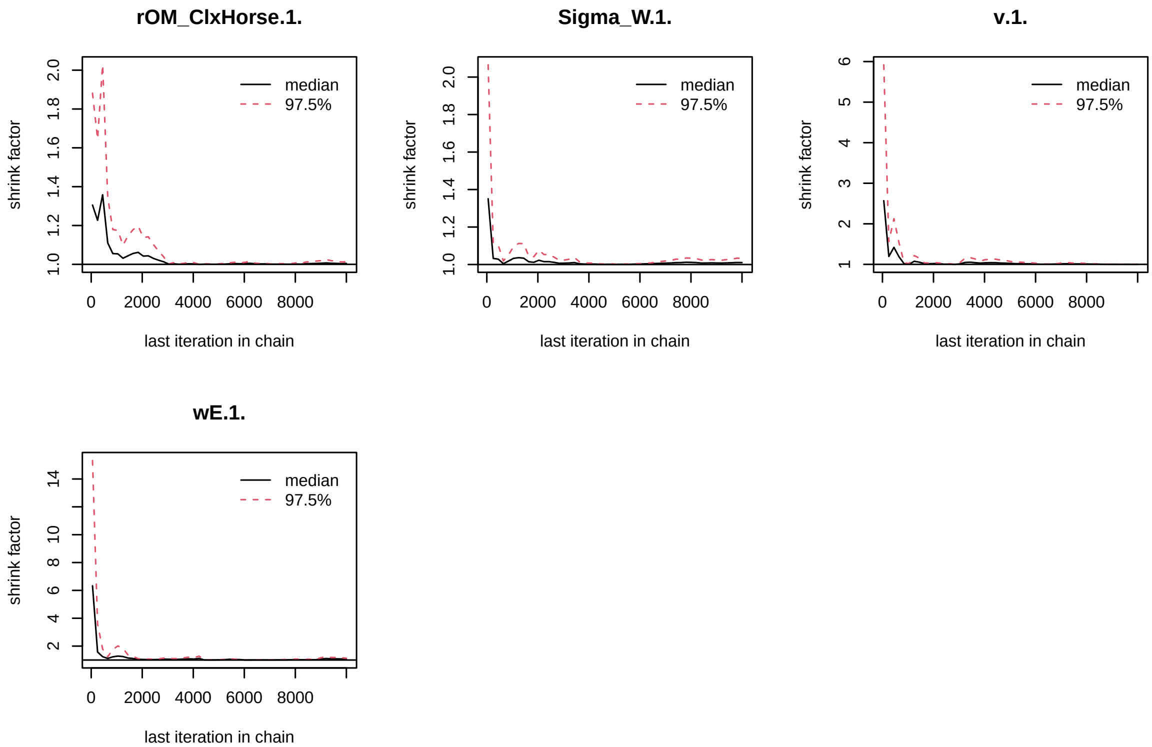


Figure 4: Chains convergence.

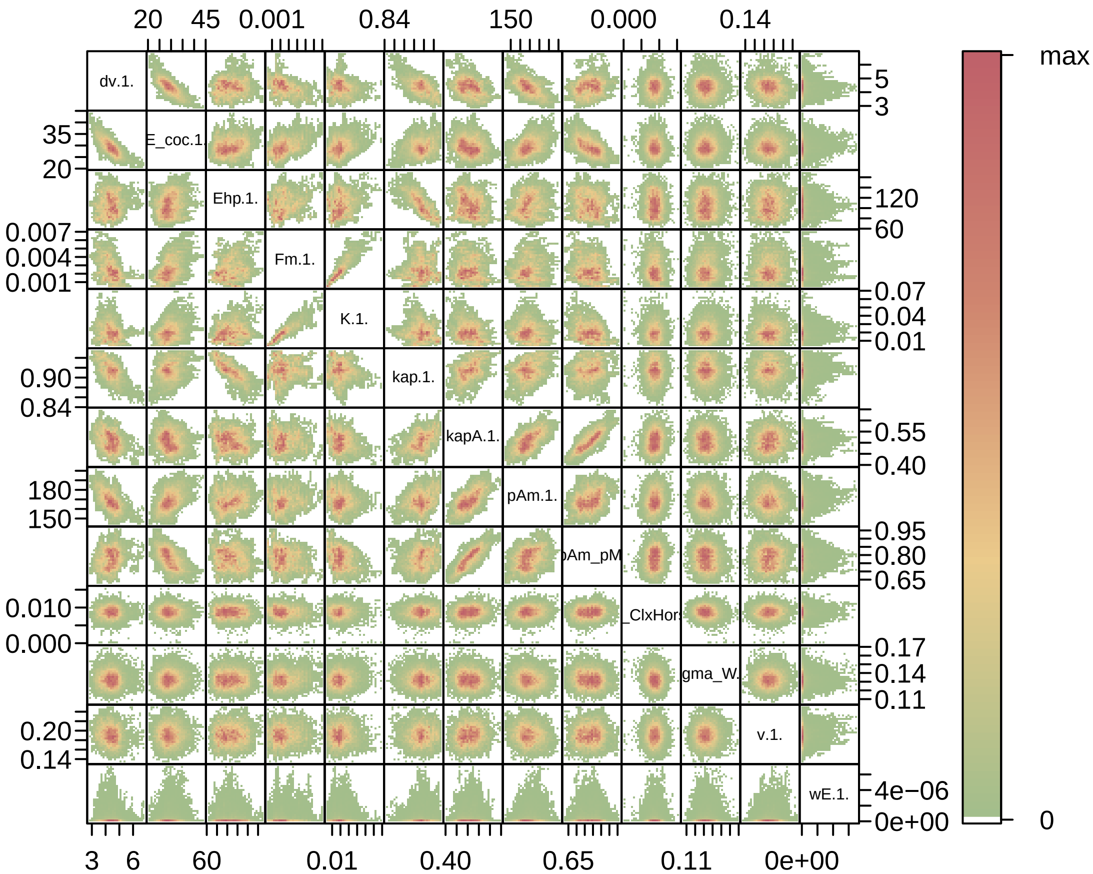


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

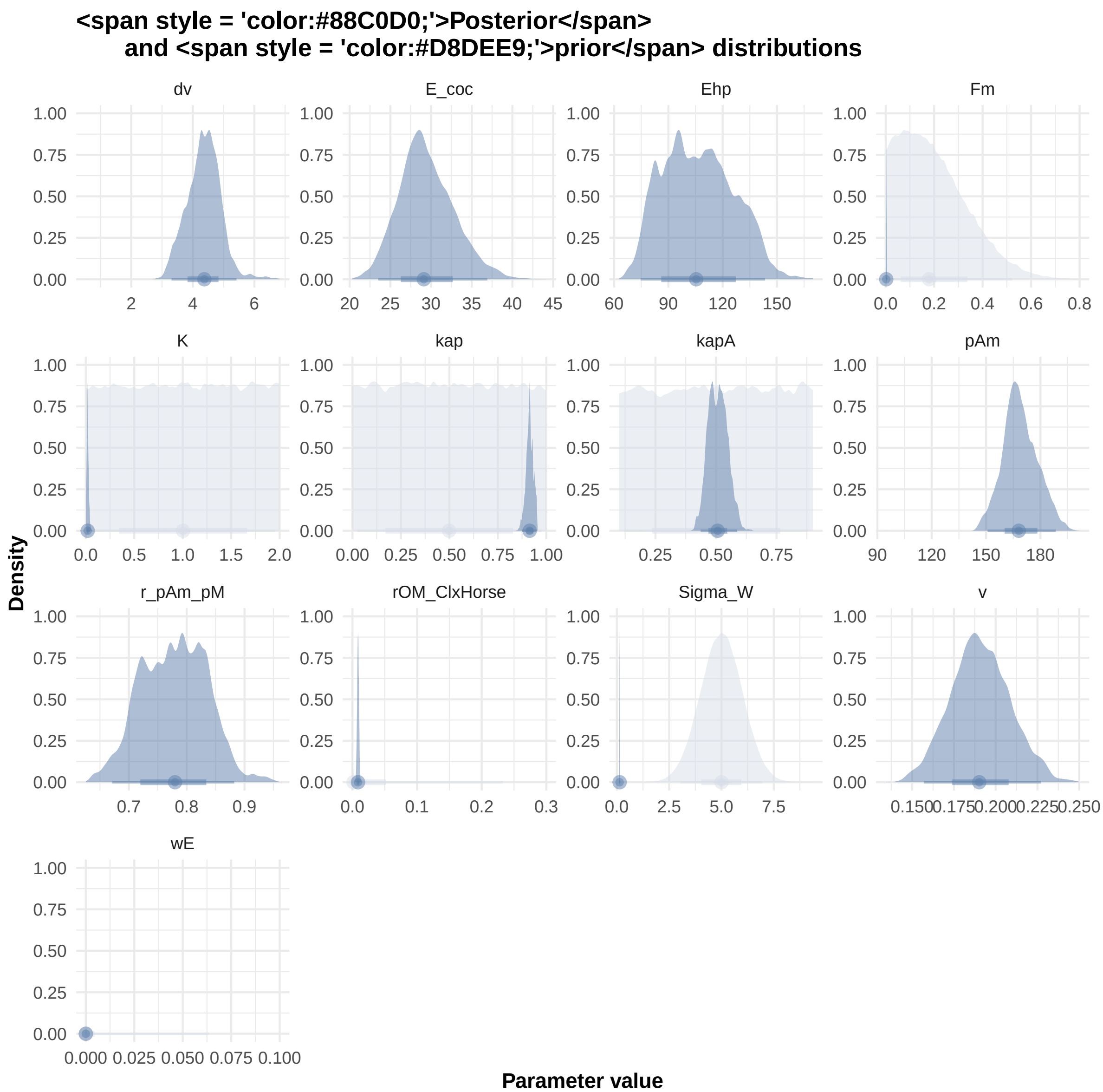


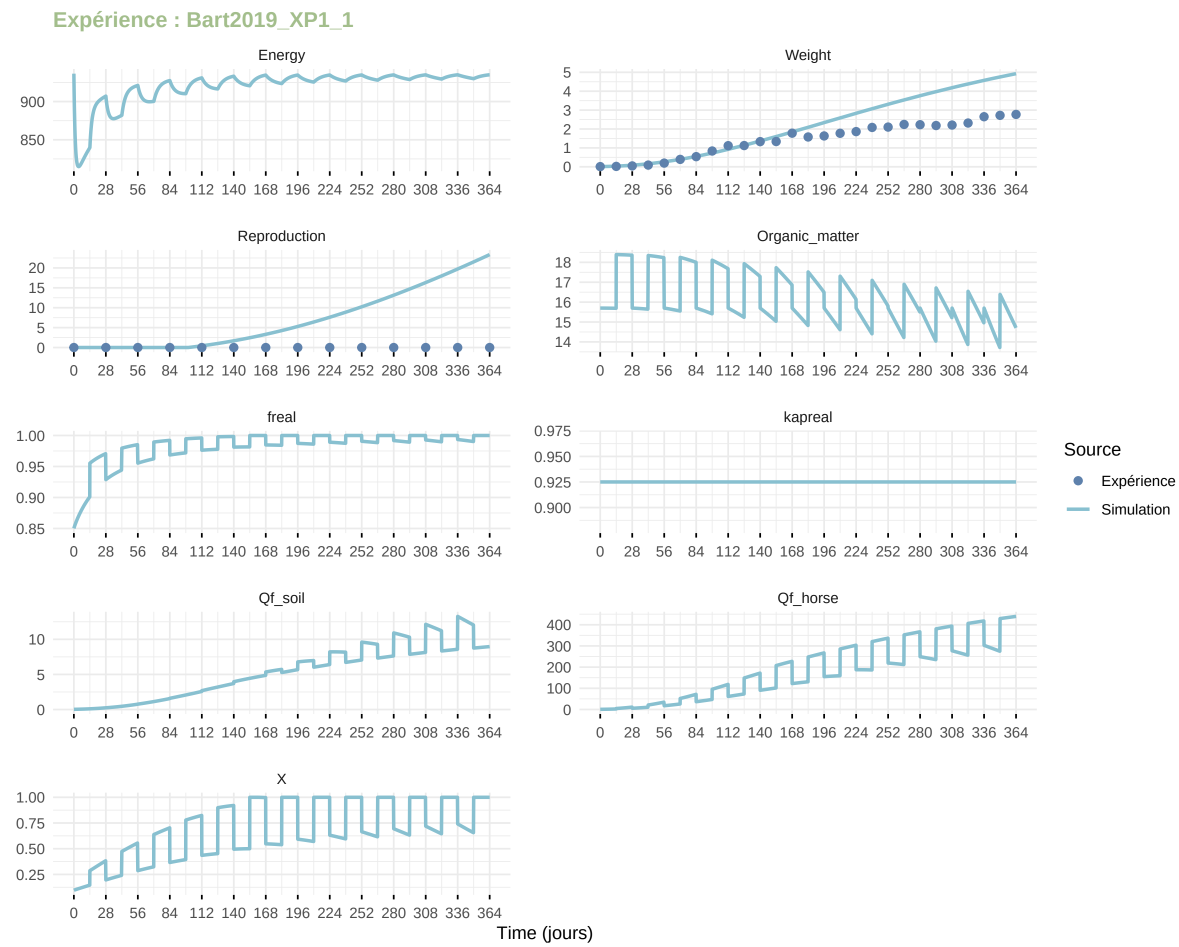
Figure 6: Distributions of the priors and the posteriors.

```r
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Length_struct,
    Reproduction,
    Energy,
    Organic_matter,
    kapreal,
    freal,
    Qf_soil,
    Qf_horse,
    X"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcali DEB_sim_full.in
```

```
##############################
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
```

```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
```

```
"many-to-many"` to silence this warning.
```



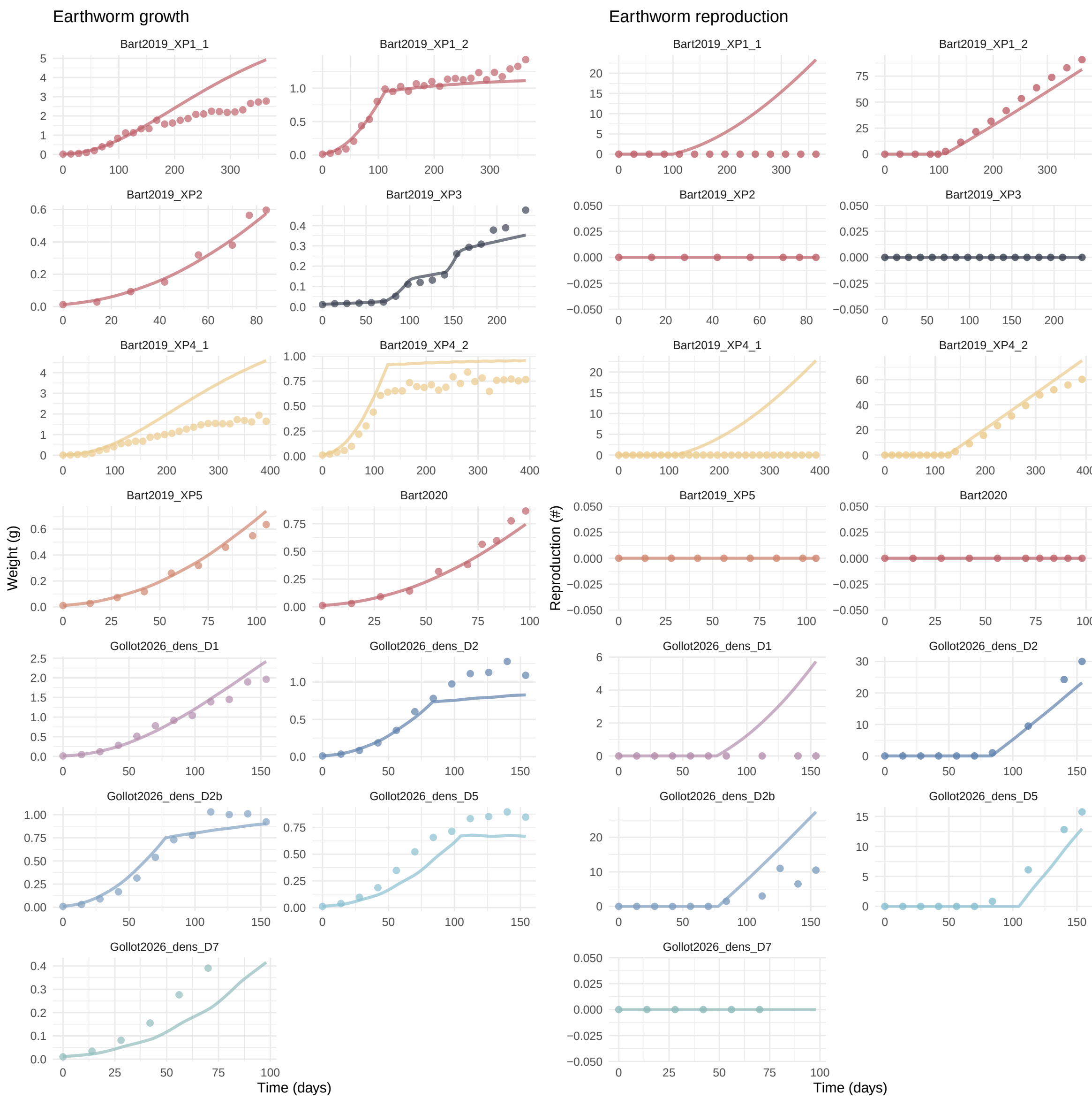
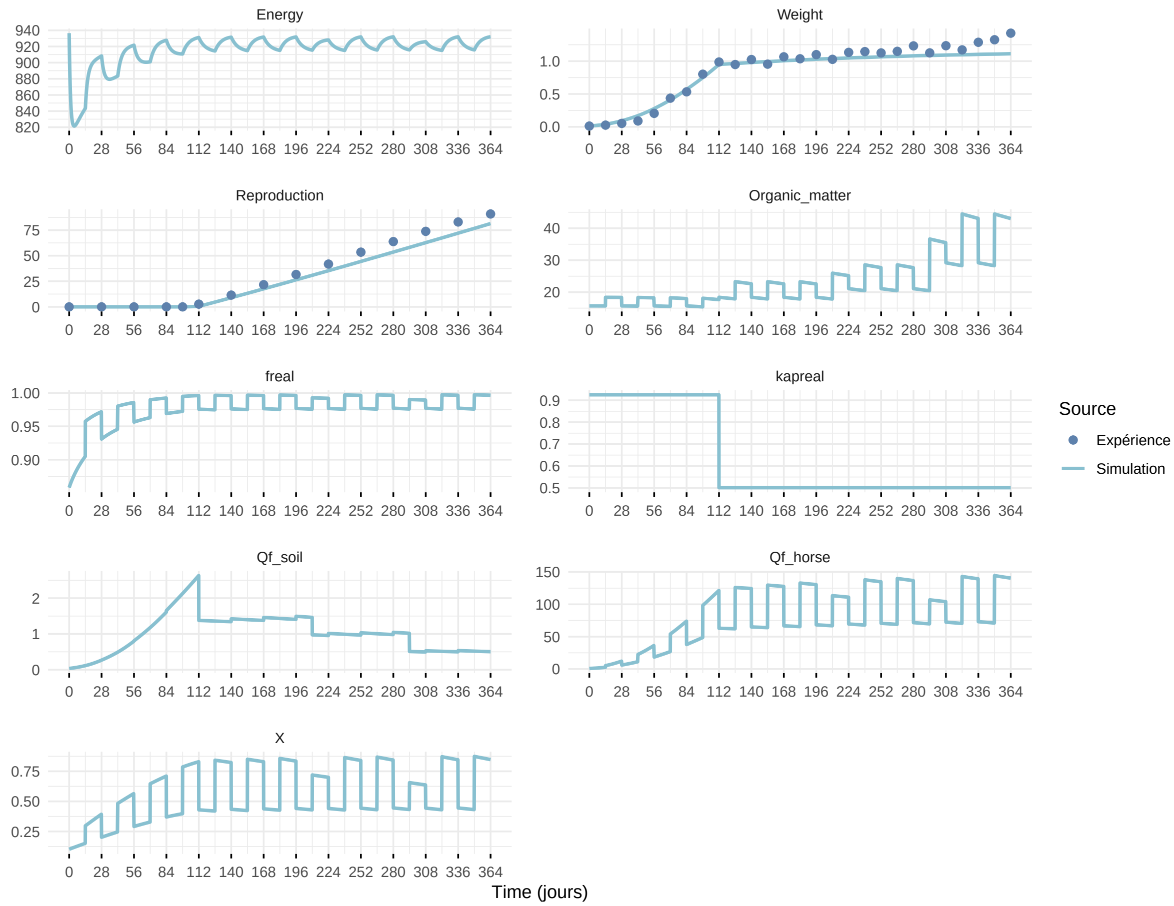
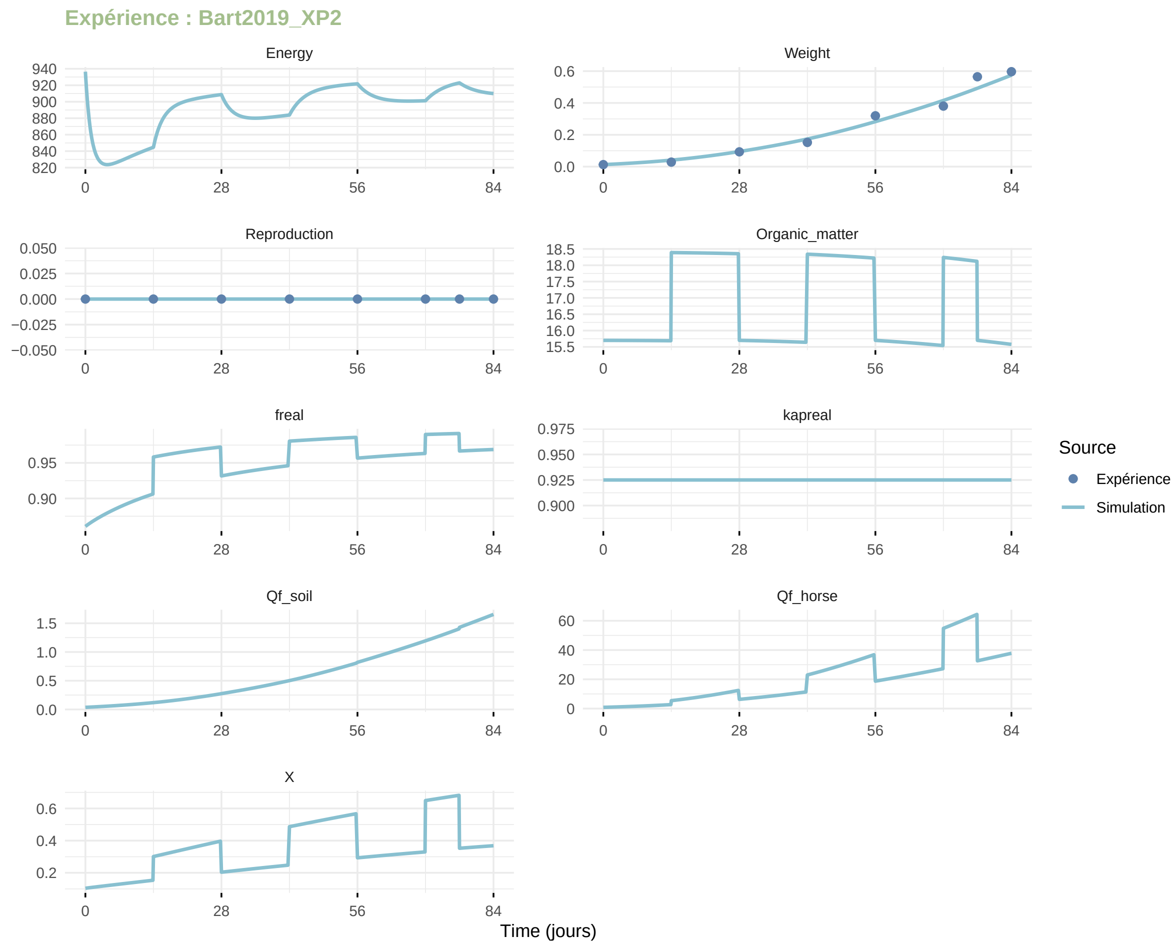


Figure 7: Simulations Weight and Reproduction.

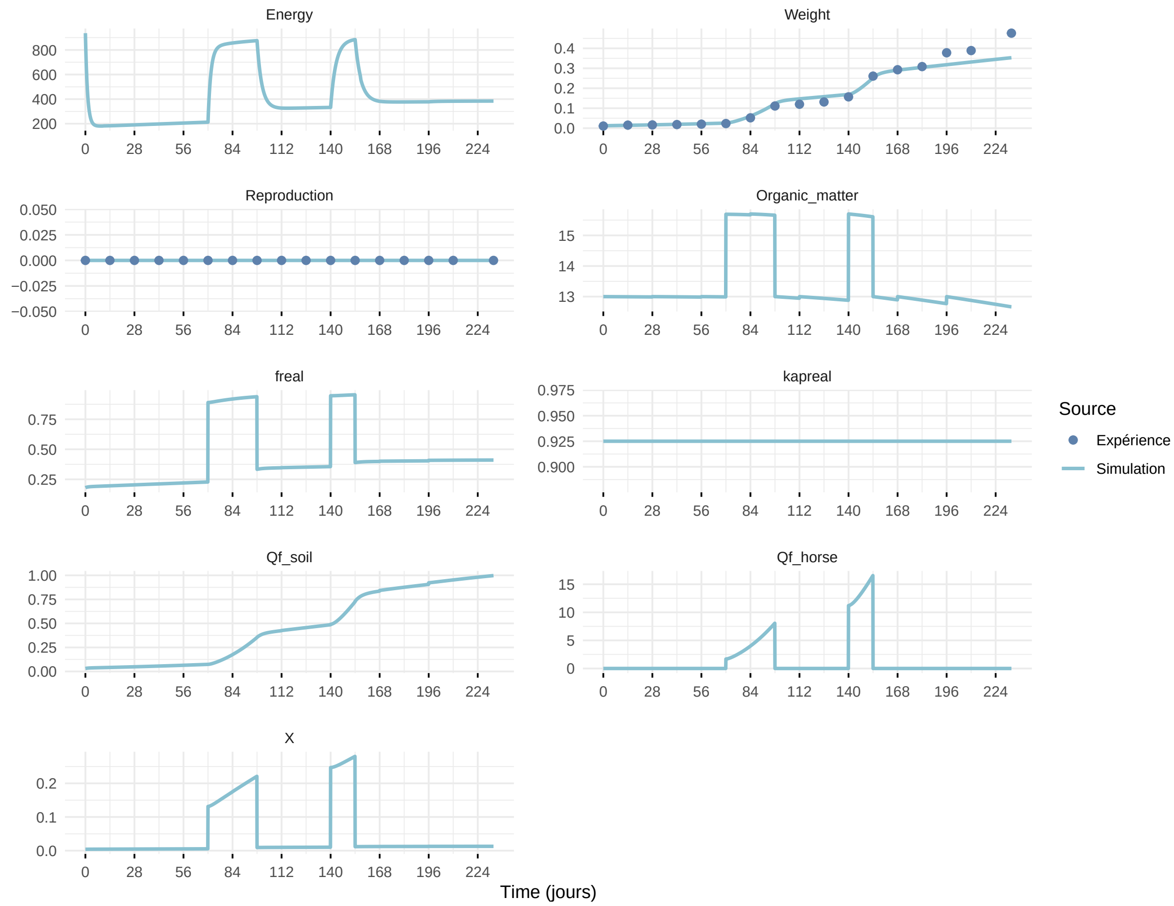
Expérience : Bart2019_XP1_2



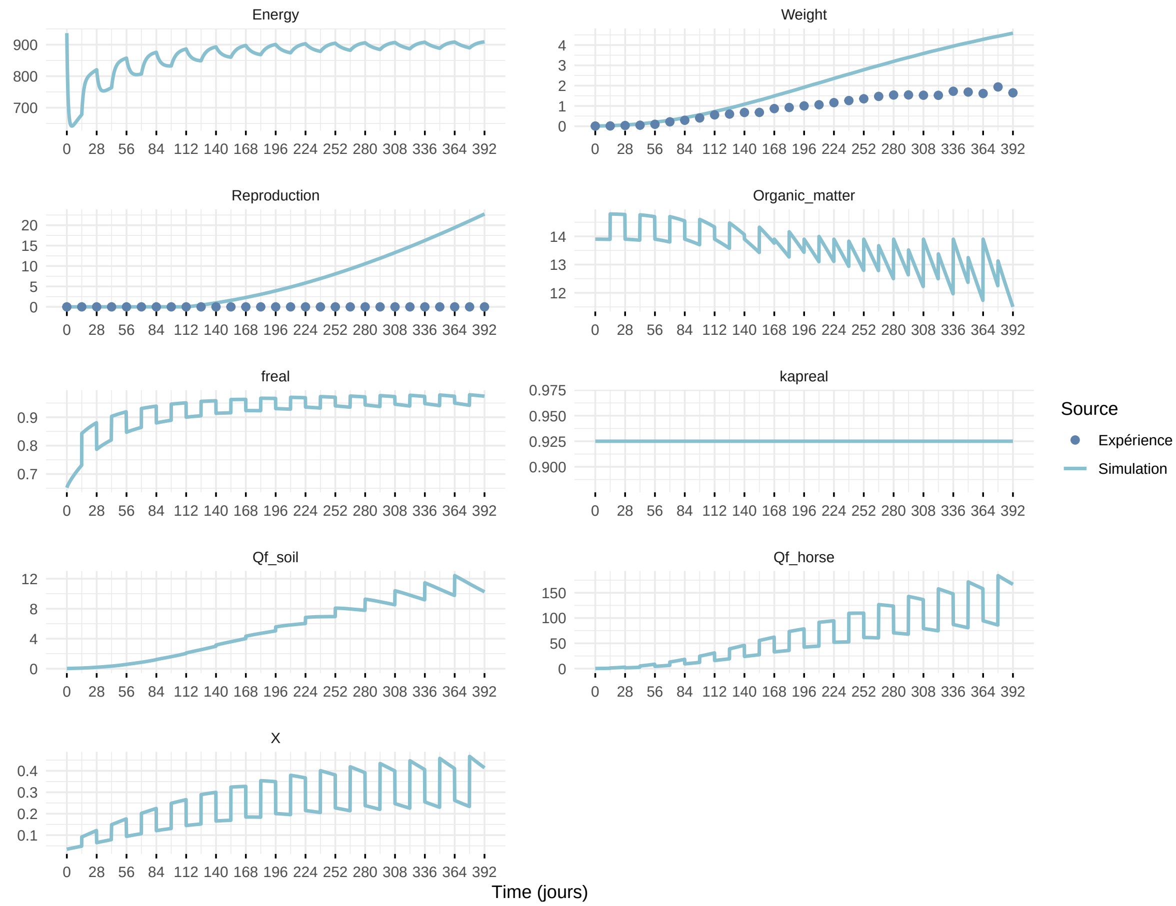


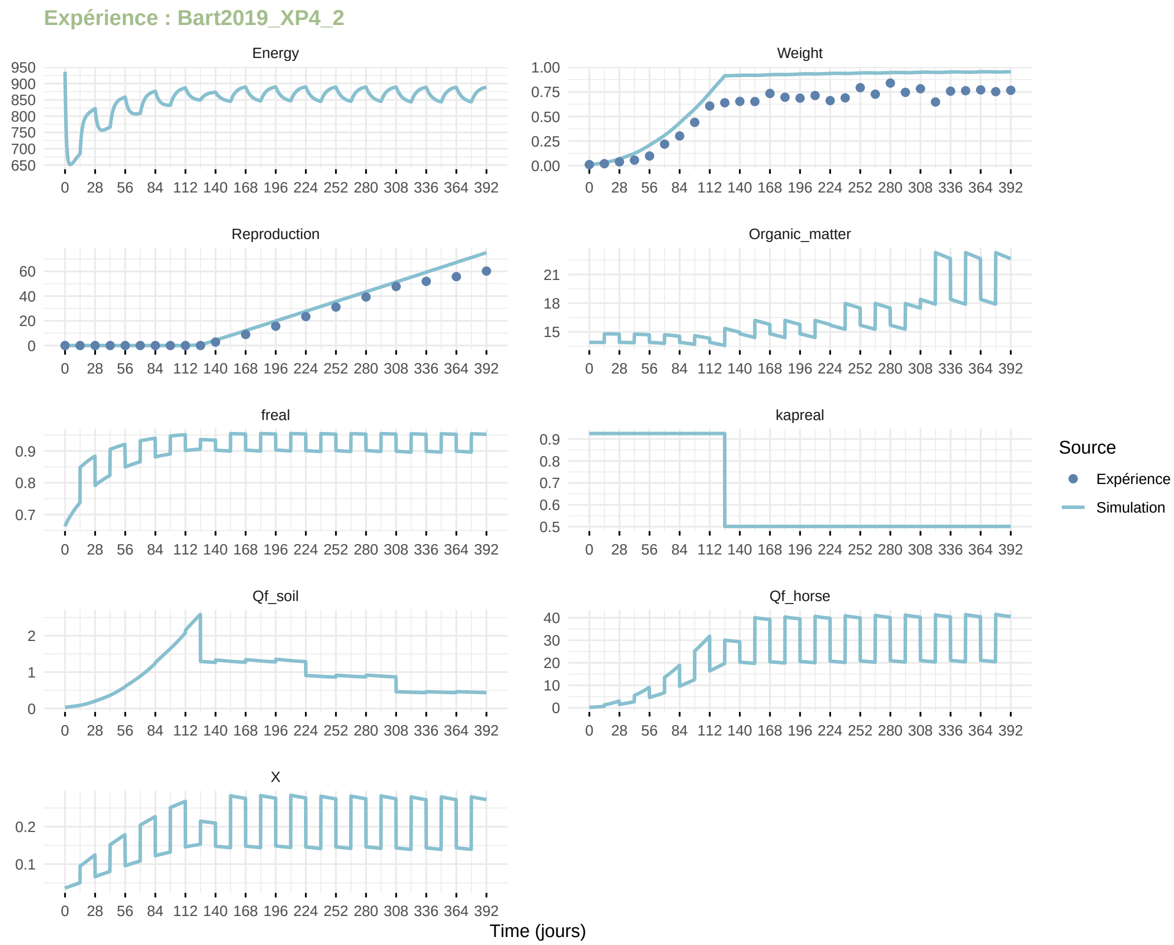
Expérience : Bart2019_XP2
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Time (jours)
Source
Expérience
Simulation

Expérience : Bart2019_XP3



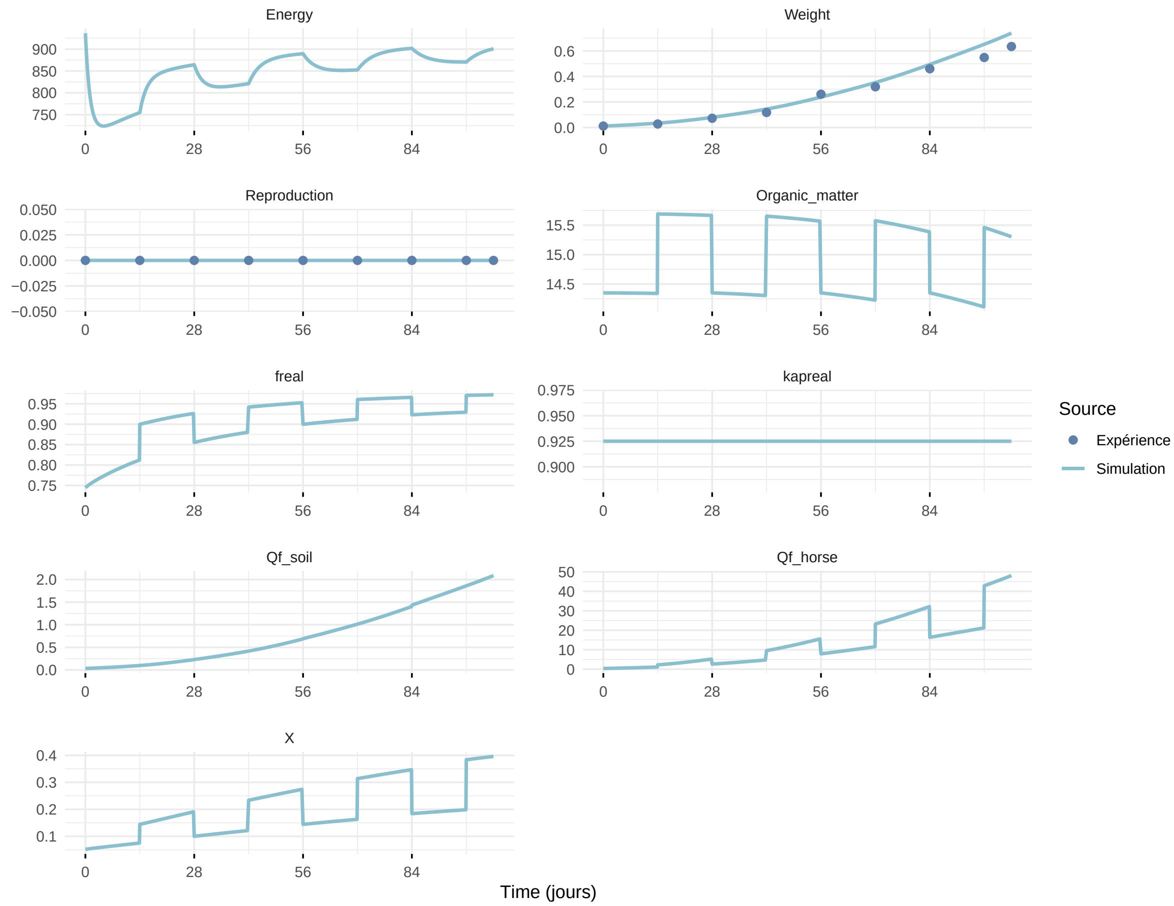
Expérience : Bart2019_XP4_1

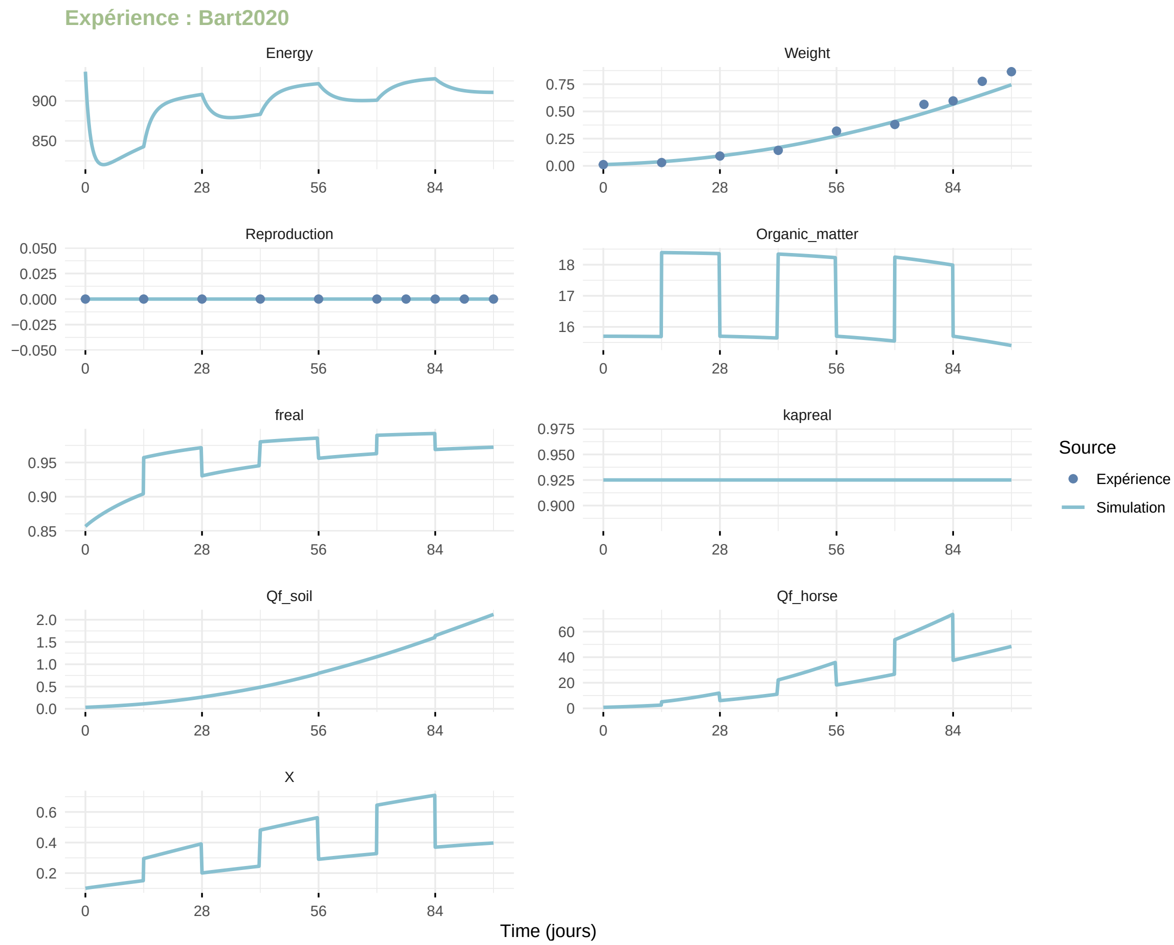




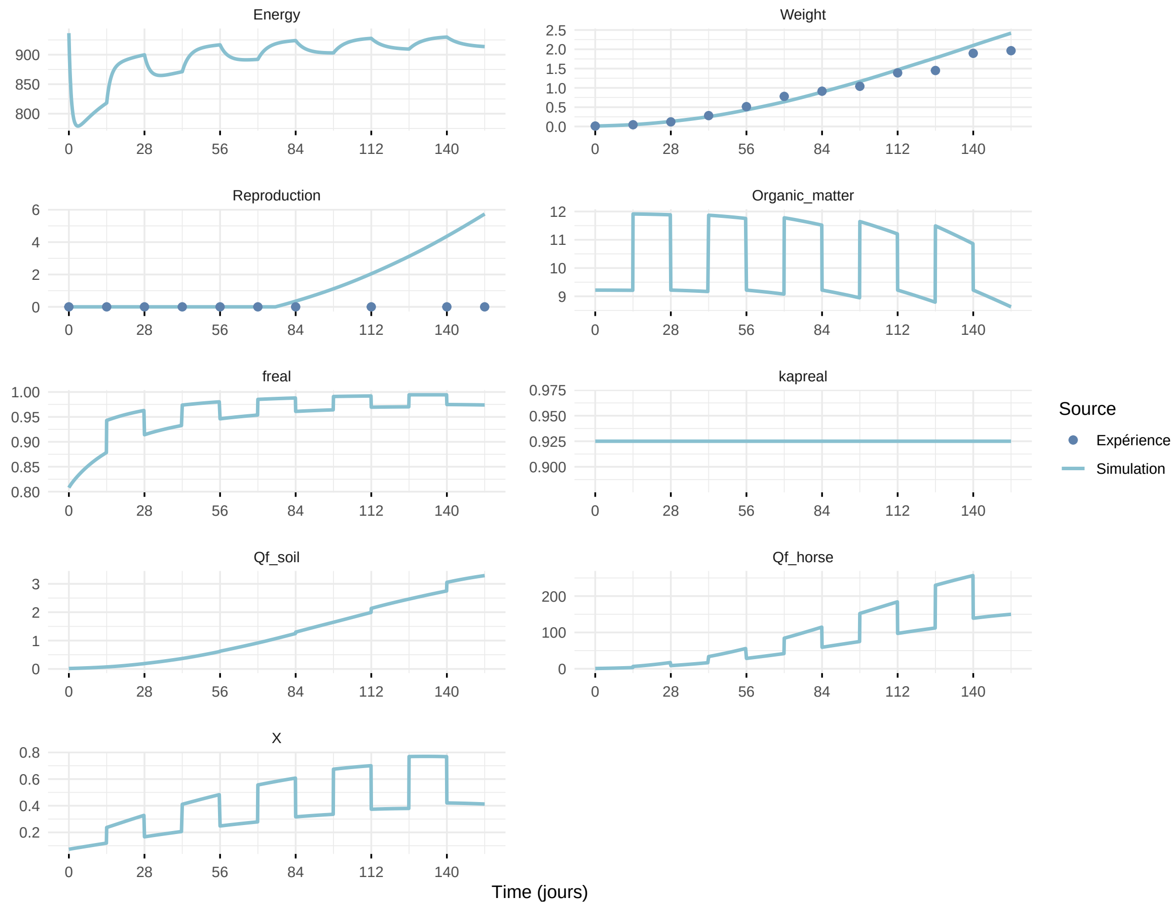
Expérience : Bart2019_XP4_2
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)

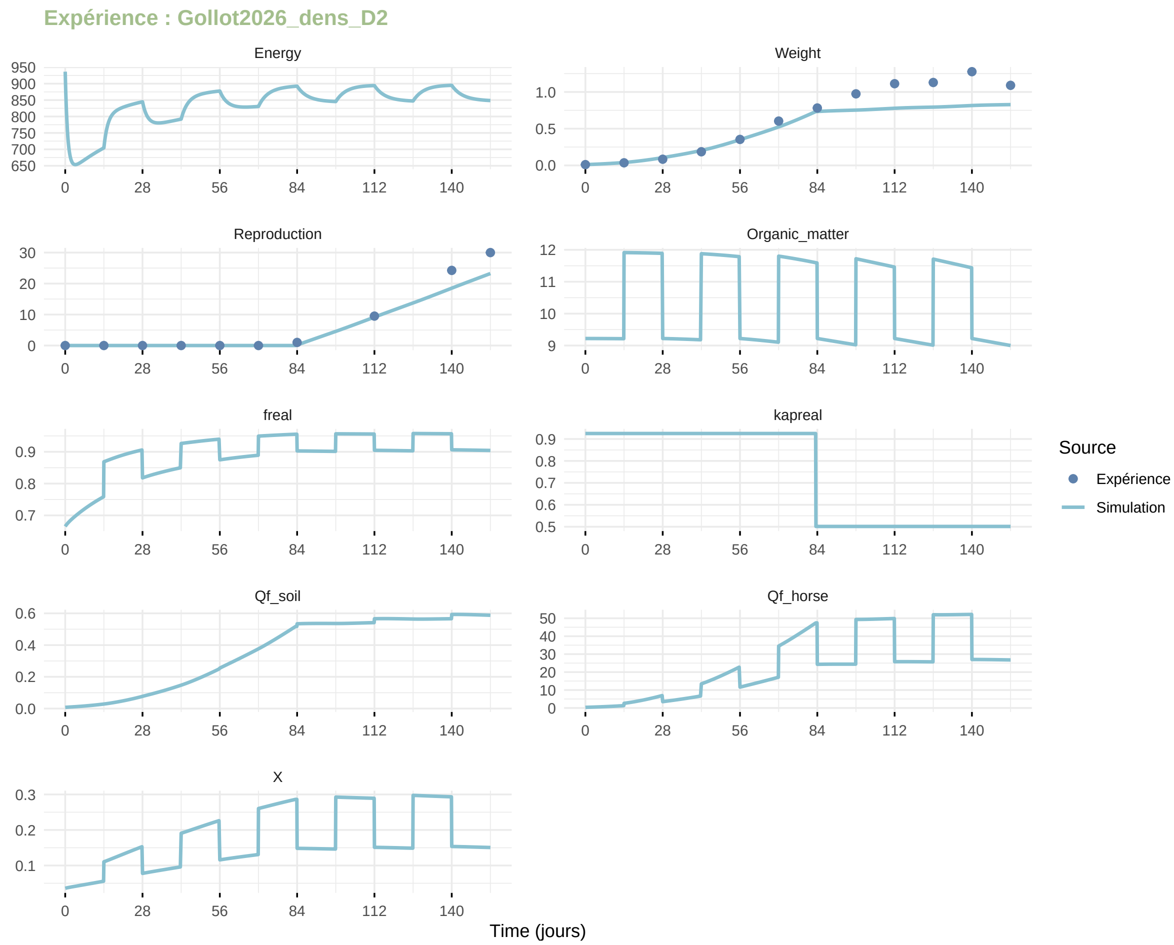
Expérience : Bart2019_XP5




Expérience : Bart2020
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)

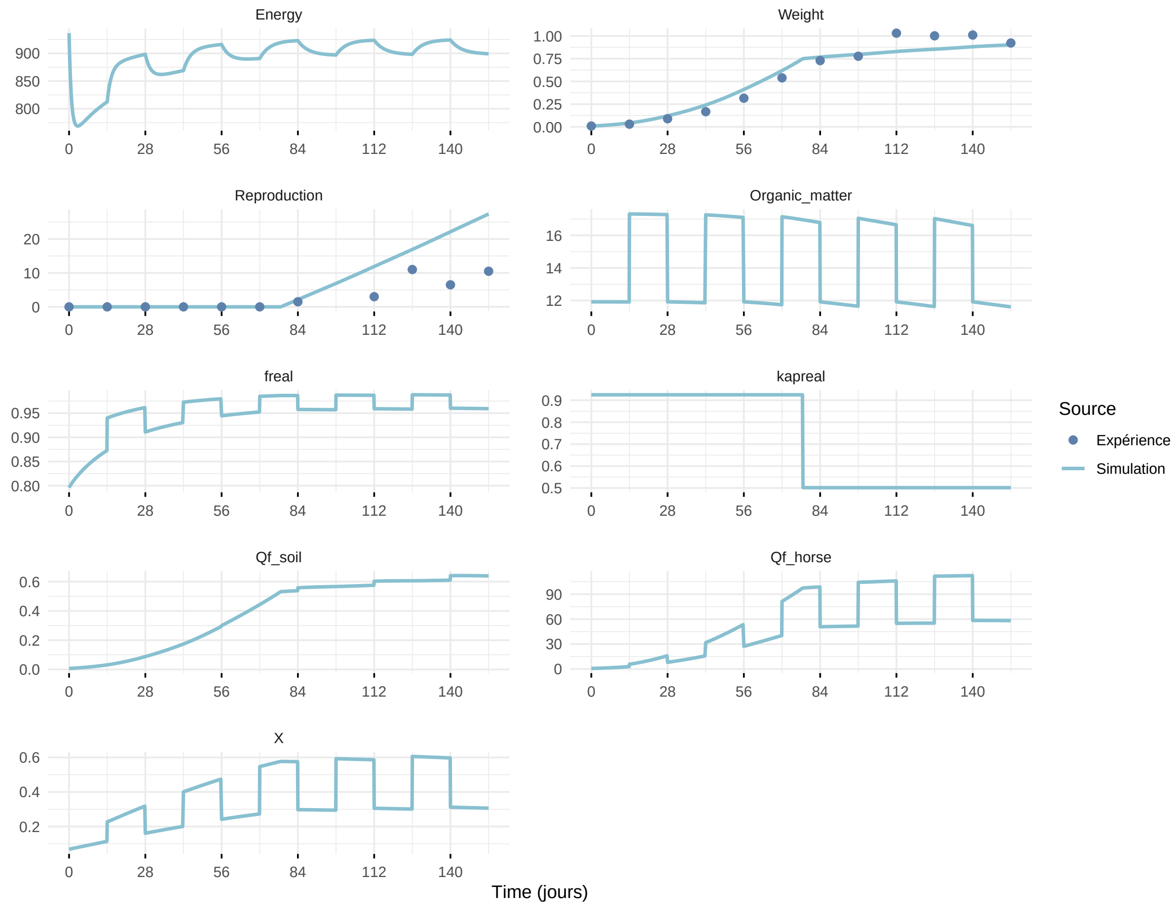
Expérience : Gollot2026_dens_D1



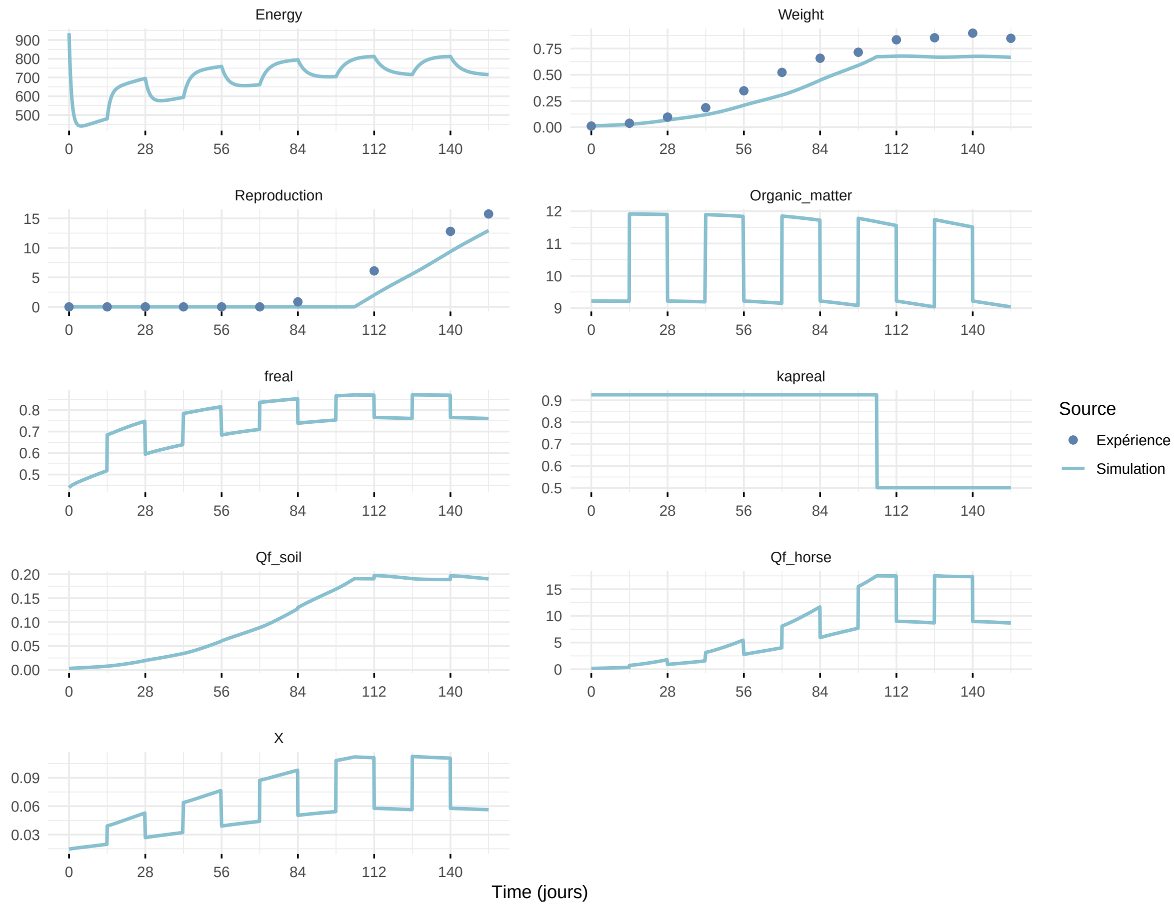


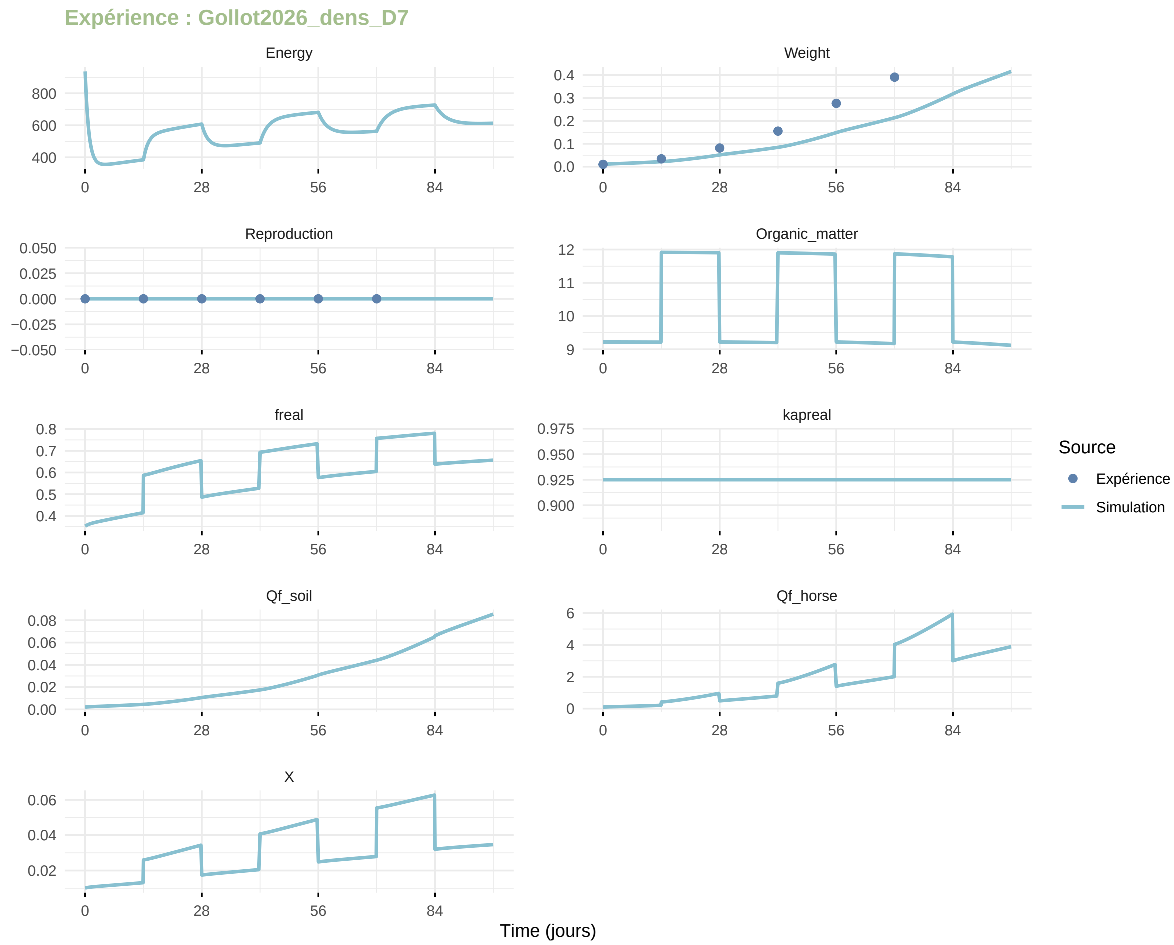
Expérience : Gollot2026_dens_D2
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Time (jours)
Source
Expérience
Simulation

Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5




Expérience : Gollot2026_dens_D7
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)